

EU SAE Dashboard

Download and Setup Guide

Updated for the current EU_SAE_application_package dashboard: run labels, upload-based data inputs, benchmark target database, population weights, archived output folders, and the Run Setup section in final_report.html.

Overview

This guide explains how to download the EU Small Area Estimation Dashboard package from GitHub and run it on your computer. The dashboard runs locally: your microdata, uploaded files, run logs, and generated outputs stay on your computer unless you choose to share them.

Preliminary release. Treat this package as a working release. Review outputs carefully before using results for official production.

Prerequisite

Install R 4.2.0 or later before running the dashboard.

Download R from: <https://cran.r-project.org/>

During installation, accept the default settings. RStudio is not required.

Step 1: Download the Package

Open the GitHub repository or releases page:

https://github.com/Noboyoshida227/EU_SAE_application_package

https://github.com/Noboyoshida227/EU_SAE_application_package/releases

Download Source code (zip). On Windows, use the .zip file. On Mac or Linux, .zip or .tar.gz both work.

Step 2: Unblock the ZIP File (Windows Only)

Do this before extracting the ZIP file. It prevents Windows from blocking the launcher script.

- Right-click the downloaded .zip file.
- Click Properties.
- On the General tab, check Unblock if the checkbox appears.
- Click OK.

If Windows shows a security warning later, right-click Start_Dashboard.bat, open Properties, check Unblock, and try again.

Step 3: Extract the ZIP File

- Right-click the ZIP file and choose Extract All.
- Choose a convenient folder, such as Documents.
- Open the extracted EU_SAE_application_package folder.

You should see files and folders such as:

app.R, app_support.R, Start_Dashboard.bat, Start_Dashboard.sh, install_packages.R, report.Rmd, R/, scripts/, sample_data/, outputs/, instructions/, www/

Step 4: Prepare Your Data

You do not need to copy private data into the package before launching. The dashboard provides file upload controls. The included `sample_data/` folder contains example files for testing.

Input	Typical file	What it should contain
Survey data	.rds	Household records with year, domain, PSU, weight, household size, welfare, poverty line or poverty inputs, and optional strata or benchmark-level column.
Auxiliary data	.rds	Domain-level covariates by domain and year. Domain and year values must match the analysis setup.
Geometry data	.rds	An sf object with polygon boundaries. Domain IDs must match the survey/auxiliary domain IDs.
Benchmark target database	.rds, .csv, .xlsx	Optional. Benchmark targets by year and, if grouped benchmarking is used, by benchmark level.
Domain population file	.rds, .csv, .xlsx	Optional. Domain populations by domain and year. If omitted, the app can estimate populations from the survey.

Population weights

For direct poverty rates and mean welfare, the dashboard uses $\text{population_weight} = \text{weight} * \text{household size}$ when the household-size column is mapped. If no domain population file is uploaded, benchmarking can estimate domain populations from the survey as the sum of $\text{weight} * \text{household size}$ by domain and year.

Benchmarking setup

If Apply benchmarking is checked, first optionally upload a Benchmark Target Database. If no target database is uploaded, the app estimates benchmark targets from the survey direct estimates. Leave the benchmark-level mapping blank for national benchmarking. Fill it with any survey column name, such as region, NUTS2, or voivodeship, for grouped benchmarking.

Step 5: Launch the Dashboard

Windows

Double-click `Start_Dashboard.bat`.

Mac / Linux

Open Terminal, navigate to the extracted folder, and run:

```
bash Start_Dashboard.sh
```

On first run, the launcher installs missing R packages. This can take several minutes and requires an internet connection. Keep the launcher or terminal window open while using the app.

The app usually opens at `http://127.0.0.1:7777`. If that port is already in use, it automatically tries the next available port, such as `http://127.0.0.1:7778`. Use the URL printed in the launcher window.

Step 6: Check Readiness, Then Run Analysis

Use Check Data Readiness before Run Analysis. This checks requested years, variable mappings, domain alignment, missing domain-year estimates, benchmark-level requirements, and population-weight readiness before the longer model run starts.

Common variable mappings include:

Mapping field	Purpose
year	Two analysis years, entered as comma-separated values in the sidebar.
domain	Small-area domain ID. Keep leading zeroes by using a string/text column if needed.
psu	Primary sampling unit for survey-design variance estimation.
strata	Optional strata ID for survey-design variance estimation. Leave blank if unavailable.
weight	Household survey sampling weight.
household size	Used with weight to create <code>population_weight = weight * household size</code> .
benchmark level	Optional grouped benchmark level. Leave blank for national benchmarking.
welfare / poverty line	Used to compute poverty FGT indicators or mean welfare.

Outputs and run labels

Each run writes current outputs to `outputs/`. The Run label field appends a readable label to the archived run folder under `app_runs/`. The Status tab shows the current file path and the saved archived copy for each expected output.

Examples:

```
outputs/final_report.html
outputs/data/pov_fh.xlsx
outputs/data/pov_mfh.xlsx
outputs/data/pov_comparison_detailed.xlsx
outputs/data/statistical_significance_comparison.xlsx
app_runs/20260623_222755_default_setup/app_config.yml
```

Understanding Section 2 Run Setup in final_report.html

Near the top of `final_report.html`, Section 2 Run Setup is the audit trail for the run. Use it to confirm the run ID and run label, locate the current outputs and archived `app_runs` folder, see whether benchmarking was off or enabled, identify the benchmark source and level, check any Benchmark Target Database file used, and confirm the selected MFH model used in the report.

Before sharing the repository through GitHub, keep `app_runs/` and generated `outputs/` files out of the shared package. The repository keeps only empty output folders and `.gitkeep` placeholders.

Troubleshooting

Problem	What to do
R is not installed	Install R 4.2.0 or later from https://cran.r-project.org/ and try again.
Security warning on Windows	Unblock the ZIP file or <code>Start_Dashboard.bat</code> from Properties.
Package installation fails	Check internet access and rerun <code>Start_Dashboard.bat</code> .
Dashboard does not open	Open the URL printed in the launcher window. It may be <code>127.0.0.1:7777</code> or a later port.
Wrong years or missing output	Run Check Data Readiness and review requested-year errors before running the model.
Benchmark-level error	Leave benchmark level blank for national benchmarking, or map a valid grouped benchmark column.
Population/benchmark warning	Map household size correctly or upload a domain population file.
Maps are blank	Check that geometry domain IDs match your data and the mapped shapefile domain column.